

George Mason University

Smarter Housing Prices Through Data

A Statistical and NLP Approach to Analyzing Housing Listing Descriptions

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1. Introduction

1.1 Background and Motivation

The factors that have been used for predicting housing prices include the square feet of the property, the number of bathrooms, location, and home age. The reasons for considering those factors is that they provide us with details about the physical nature of the property.

Nevertheless, in addition to those details, real estate listings also contain some text in terms of description.

Words such as luxury, modern, and renovated are among the common descriptive words that might influence the perception of potential buyers of the properties and also indicate certain characteristics of houses. Therefore, we may consider whether the listing descriptions could help us to predict housing prices and which words would correspond to higher or lower prices. And this is thanks to advanced NLP techniques, we will be able to turn text into numeric data.

1.2 Research Question and Objectives

The main research question for this project is:

Do listing descriptions influence housing prices beyond structural housing features?

To answer this question, this project compares models with and without text features. Traditional housing variables such as square footage, bathrooms, and home age are used along with text features created from listing descriptions.

The main goals of this project are:

- See whether text improves model performance
- Find words connected to higher or lower prices
- Compare regression and machine learning models
- Test whether text still matters after controlling for housing features

The overall goal is to understand whether listing descriptions contain useful information about housing prices.

2. Literature Reviews

2.1. Traditional Housing Price Models

The conventional techniques for house price estimation largely depend on structured variables like the size of the house, number of bedrooms, number of bathrooms, location, and age of the

house among others. The most popular technique in this category is hedonic pricing, where analysis of housing attributes and their impact on price is carried out.

This technique is beneficial since it makes interpretation of results simple, thus giving an understanding of the impact of different housing attributes on prices. However, traditional methods normally overlook unstructured data like the house description, hence failing to capture significant attributes that can contribute to the price of the house.

2.2 NLP & Text Mining in Real Estate

From recent literature in this field, various studies have adopted NLP and text mining methodologies in order to analyze textual description of real houses that are on sale. In essence, text mining techniques create a chance where textual information can be translated to numeric data which can be applied in machine learning.

According to previous research, the application of terms such as "luxurious," "new build," and "waterfront" results in higher prices of houses as compared to when such terms as "auction" or "buildable" are applied. There is further evidence showing that the combination of textual data with conventional data enhances the predictive power of the algorithms.

TF-IDF, Word2Vec, and Deep Learning are becoming popular in real estate analysis owing to their ability to capture implicit information from textual data.

2.3 Research Gaps

Even though the previous literature proves that adding text can enhance the ability to predict house prices, most of the papers concentrate solely on prediction and ignore the statistical explanation part. The papers apply complex deep learning methods or do not even make a comparison between models with and without text.

However, our project aims to fill some of those gaps through:

- Comparing the predictive performance of models with text features and those without
- Analyzing whether the presence of text features is significant while taking into account other factors
- Finding out what type of words can predict price increase and decrease
- Integrating both statistical and machine learning approaches

3. Dataset & Preprocessing

3.1 Data Collection & Structure

For this project, the dataset used is the housing listings in Virginia extracted from Realtor.com using the HomeHarvest, which is a Python package. At the beginning of this project, the direct web scraping approach was applied to sites like Zillow and Realtor.com, but only the first 40 listings could be accessed, making it impossible to have a sufficiently large dataset.

In order to overcome the challenge posed by the limited number of listings that could be accessed, the HomeHarvest package was adopted. Through HomeHarvest, we were able to gather a far larger number of housing listings. The total number of housing listings gathered was about 9,800 while the number of features was 66.

Some of the structured features include square footage, number of bathrooms, house age, and locations among others. The unstructured data in the dataset involved the description provided on the listing pages.

```
from homeharvest import scrape_property
from datetime import datetime
import pandas as pd

current_timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")

listings = scrape_property(
    location="Virginia",
    listing_type="for_sale",
    past_days=30
)

print(listings.shape)
print(listings.head())

listings.to_csv(f"fairfax_homeharvest_{current_timestamp}.csv", index=False)

(9800, 66)
```

3.2 Cleaning & Missing Data

As we know, real world data is often messy and not fit for modeling. The first things that data scientists often check for is missing and duplicate data. The original dataset contained many columns that had more than 80% of its data missing so those columns were removed entirely and so were any duplicate data. Features that also did not pertain to the project such as 'broker_id', 'property_url', 'agent_email', etc., were removed. Missing data in features that were important to the project were filled with either the median for numerical features and "Unknown" for categorical features.

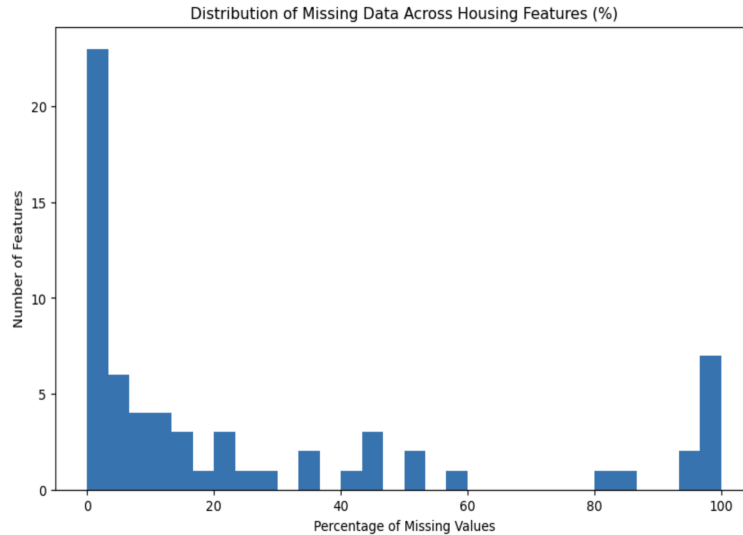


Figure 1: Percent of missing values in the original, un-cleaned dataset illustrating a large number of features with more than 80% missing features.

3.3 Feature Engineering and Multicollinearity

In order to enhance model performance, several new features were engineered, including combining features with similar information content. For instance, instead of having two separate features for full and half bathrooms, the number of bathrooms was summarized as total baths. Combining highly similar features not only reduced the number of features we were working with but also decreased the multicollinearity between features. Along with combining features, log transformation was utilized for heavily skewed features which helped balance and improve model performance.

Additionally, several text-based features were engineered, based on patterns found in real estate listing descriptions. The goal of these features was to capture typical topics mentioned by real estate listing descriptions.

- The premium score feature counted the frequency of luxury-related words and features such as granite countertops, stainless steel appliances, hardwood flooring, etc. Premium score features are usually associated with higher-end properties.
- The layout score counted words such as open concept, primary suite, or home office. Buyers might prefer homes with a specific design.

- Condition score counted words related to the general condition of the property, such as renovated, updated kitchen, new construction, etc.
- The outdoor score counted the occurrence of words related to exterior features, such as deck, covered porch, walking trails, etc.
- Location score counted location-related words such as waterfront, mountain views, prime location, etc.

These features helped convert listing descriptions into numerical variables that could be used in the models. They also helped summarize pretty important information found in the text in a simpler and a way more organized way.

3.4 Text Processing & TF-IDF

Pre-processing of the textual descriptions was performed prior to running models. Words were made lowercase, numbers were stripped away, and any unnecessary whitespace was pretty much ‘cleaned up’. The end result was an increased consistency in the text to perform further analysis on.

Following pre-processing, TF-IDF was applied to turn these descriptions into feature vectors. The TF-IDF value represents the importance of a given word in a document relative to other documents.

4. Methodology & Models

4.1 Train/Test Split

To ensure that there was an objective assessment of the performance of the models, the data set was divided into two sections: the training and testing data sets. The data set was split such that 80 percent of the data was used for training purposes while the remaining 20 percent was used to evaluate the models. A static random state was applied during this process too.

4.2 Regression Models

Regression models were initially used as a benchmark technique to predict housing prices. These models included common features such as house size, number of bathrooms, house age, and location.

Text features like premium score, condition score, and layout score were incorporated into regression models to determine whether housing description data improves the prediction

accuracy of the model.

The use of regression models was important since it is easily interpretable and testable statistically. The coefficients can be used to explain how different features affect housing prices.

4.3 TF-IDF + Ridge Regression

One of the main models use in this project was TF-IDF combined with Ridge regression. TF-IDF was used to convert listing descriptions into numerical features based on how important certain words are across all listings. This allowed the model to learn patterns from the text descriptions. Because TF-IDF creates a very large number of features, Ridge regression was used to reduce overfitting and improve model stability. Ridge regression helps prevent the model from relying too heavily on any single word or feature. This became the best-performing model in the project and showed that text descriptions significantly improved housing price prediction.

4.4 Decision Tree Models

Also, decision trees were considered for determining how the data could be segmented into various price ranges by applying decision rules.

Decision tree models differ from regression in their ability to accommodate nonlinear relationships among variables. The results revealed how attributes like the size of the house, the number of bathrooms, as well as certain text-based attributes, assisted in classifying houses into high and low-price categories.

Two types of decision tree models were compared:

- A decision tree model based on structural attributes only
- A decision tree model based on structural and text-based attributes

4.5 Grid Search & Evaluation Metrics

The Grid Search method was applied to test various combinations of hyperparameters and choose the best one for each model. Different combinations of alpha value for the Ridge algorithm and TF-IDF configuration have been tested to increase the performance.

A few methods of assessment were employed in order to evaluate the performance of the models:

- R^2 : measures how much variance in housing prices is explained by the model

- RMSE: measures the average value of errors made in predictions
- MAE: measures the average absolute error in predictions

Train vs Test performance comparison was also carried out to detect possible issues with overfitting.

5. Results & Discussions

5.1 Model Comparison

The aim of the comparison was to see which models perform better in predicting housing price data and how much text-based features help in increasing their effectiveness. The model that did not include any text-based feature gave an R^2 score of approximately 0.58. Upon introducing the text-based features, there was a great increase in performance. The most effective model had the name TF-IDF + Ridge and gave an R^2 score of about 0.82.

Model	Training RMSE	Testing RMSE	Training R^2	Testing R^2
Baseline Regression (No Text)	0.609	0.609	0.577	0.577
TF-IDF + Ridge	0.335	0.480	0.853	0.737
Tuned TF-IDF + Ridge	0.214	0.400	0.817	0.817
Decision Tree (No Text)	NA	0.684	0.563	0.466
Decision Tree (With Text)	NA	0.660	0.503	0.503

Figure 2: Comparison of model performance across training and testing sets

5.2 Regression & Decision Tree Results

It was shown that several structural parameters significantly affect the prices on the housing market. In particular, square footage, number of bathrooms were important determinants for prices. Some text-based parameters such as premium score, condition score, and layout score

were statistically significant as well.

The most accurate results were provided by the TF-IDF + Ridge model. This approach allowed the learning algorithm to recognize important patterns from text and use them together with Ridge regression to prevent overfitting.

Moreover, decision tree algorithms were considered. These models allowed getting more insights about price groups division and the role played by structural features and texts. Decision tree algorithms confirmed that structural factors were important determinants for prices. Textual factors were relevant for modeling decisions to a smaller extent. The accuracy of decision tree algorithms was lower compared to TF-IDF + Ridge model.

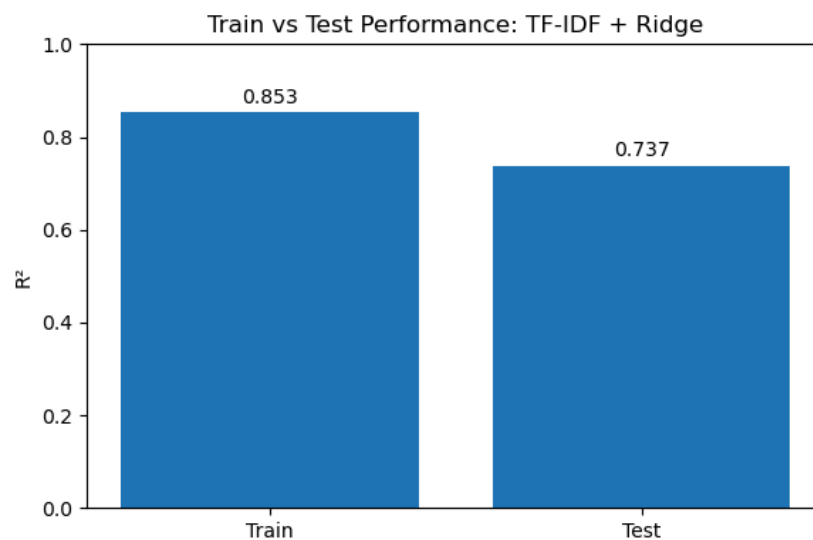


Figure 3: Training and testing R^2 scores for the TF-IDF + Ridge regression model. The model resulted in a training R^2 of 0.853 and a testing R^2 of 0.737, showing strong predictive performance with relatively good generalization to unseen housing listings.

5.3 Text Impact Analysis

Among the key objectives of this project was to determine whether listing descriptions affect house pricing in addition to structural attributes. The findings revealed that text attributes play a critical role in enhancing prediction accuracy. There was significant improvement in the value of R^2 when text was introduced to the model, as opposed to models lacking text features. Certain words were found to be related to high or low prices. The terms "acres," "estate," "development," and "waterfront" tended to relate to high prices, while "auction," "harbour," and "building" words

were related to low house prices. Statistically, homes with certain words, for example "luxury," were proven to have higher prices even after accounting for housing structural characteristics. Overall, these findings indicate that there is information in text attributes regarding house quality, buyer perceptions, and price.

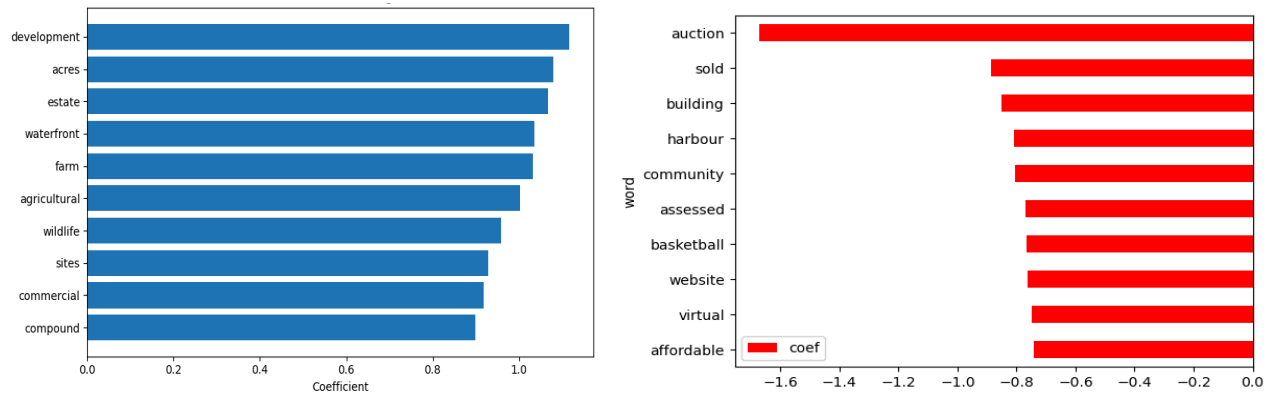


Figure 4: Model output results of certain words linked with higher (right) and lower (left) prices

5.4 Comparison to Literature

The results obtained in this study are in line with previous literature on real estate and NLP. In previous research, the authors found out that a combination of textual information together with other housing characteristics helps in building effective predictive models. Additionally, previous researches using NLP techniques have demonstrated that the value of R^2 obtained in housing price prediction models was around 0.7–0.8. In the current study, the most efficient predictive model had a value of R^2 around 0.82, which is similar or slightly better compared to previously published results.

5.5. Practical Implications

The results of this project can be applied practically in many ways, including helping to understand some aspects of real estate analyses and housing prices prediction. For people who are selling and buying houses, our research shows that the language used while describing the property may play an important role in shaping its perception in the market. Some words may even increase the buyer's interest in the product or describe the quality of the house better than

other numerical parameters do. Also, this approach can be used by real estate companies and analysts in order to make their predictions more precise.

6. Limitations & Future Work

While the developed models showed good performance results, there were also a number of limitations.

First, the data only contained the housing listings of one state and covered only a certain period of time. As time passes, markets may change, and markets may also differ from each other depending on the location. So, it is pretty unlikely that the results will be able to reflect other housing markets.

Secondly, the descriptions for the houses were created by the respective real estate agencies. Due to this, there may be some marketing terms used, incomplete descriptions, as well as some repetition of phrases in different listings. Therefore, there may be words, which may represent marketing techniques rather than an actual description of the property.

Thirdly, the project utilized TF-IDF, which mainly focuses on the frequency of use of words and does not comprehend their meaning and context in the sentence or phrase.

Moreover, a number of factors were not included in the dataset. These include the quality of local schools, crime rate, interests, economic conditions, etc.

For future research, better approaches to NLP, such as Word2Vec, BERT, or transformers in general, might be explored to improve the precision of representation of meaning within listing descriptions. Moreover, future research can make use of larger datasets and cover multiple states as well as longer periods to boost generalization power.

Finally, other machine learning algorithms like Random Forests, XGBoost, or neural networks can be applied to check if they perform better or worse than those chosen for this research.

Overall, this research revealed the existence of useful insights about house pricing in listing descriptions. Nonetheless, there is significant potential for improvement in relation to the analysis of text information in prediction models.

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